

Week 2 Module:

Date: 17/06/2026

1. Find Student Roll Number (Linear Search)

Problem Statement

Your class teacher has a list of student roll numbers stored in the order they registered. Since the list is unsorted, you need to find whether a particular student roll number exists.

Write a program to search for the given roll number using **Linear Search**.

Input Format

- First line contains integer N (number of students)
- Second line contains N roll numbers
- Third line contains integer X (roll number to search)

Output Format

- Print "Found at index i" if present
- Otherwise print "Not Found"

Constraints

- $1 \leq N \leq 1000$

Example

Input

```
5
101 105 113 110 130
110
```

Output

```
Found at index 3
```

2. Search Product ID (Binary Search)

Problem Statement

An online store keeps product IDs sorted in ascending order. A customer wants to know whether a particular product exists in inventory.

Write a program to search the product using **Binary Search**.

Input Format

- First line contains N
- Second line contains sorted product IDs
- Third line contains target ID

Output Format

Print:

Product Available

or

Product Not Available

Constraints

- $1 \leq N \leq 10^5$

Example

Input

```
6
10 20 30 40 50 60
40
```

Output

```
Product Available
```

3. Arrange Exam Scores (Bubble Sort)

Problem Statement

A teacher wants to arrange students' exam scores in ascending order to prepare the result sheet.

Write a program using **Bubble Sort**.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print the sorted array.

Example

Input

5
89 45 78 12 99

Output

12 45 78 89 99

4. Arrange Library Book Numbers (Insertion Sort)

Problem Statement

A librarian receives book serial numbers one by one and wants to place each new book in the correct position.

Write a program using **Insertion Sort** to arrange book numbers.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print sorted book numbers.

Example

Input

5
55 23 87 11 34

Output

11 23 34 55 87

5. Rank Players by Score (Selection Sort)

Problem Statement

A gaming club wants to rank players based on scores from lowest to highest.

Write a program using **Selection Sort**.

Input Format

- First line contains N
- Second line contains player scores

Output Format

Print sorted scores.

Example

Input

```
5
70 20 90 10 40
```

Output

```
10 20 40 70 90
```